Numerical Linear Algebra WS 26/27 Notes

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2026-10-21

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Preface

1 Overview

1.1 Numerical Linear Algebra Overview

1.1.1 Main computational problems

Problem	Mathematical formulation	Typical numerical methods
Linear systems	$Ax = b$	LU, Cholesky, CG, GMRES
Eigenvalue problems	$Av = \lambda v$	Power iteration, QR iteration, Lanczos, Arnoldi
Least-squares problems	$\min_x \|Ax - b\|_2$	QR, SVD
Matrix factorizations	$A = QR,\ A = LU,\ A = U\Sigma V^*$	Householder, Gaussian elimination, SVD algorithms
Low-rank approximation	$A \approx UV^*$	Truncated SVD, randomized algorithms
Matrix functions	$f(A)v$, e.g. $e^A v$	Polynomial and Krylov approximations

1.1.2 The special importance of large and sparse matrices

large and sparse matrices arise naturally in many settings, e.g. determining the stationary temperature distribution in a two-dimenstional plate.

Iterative methods and subspaces exploit sparsity. This is more effective than simply extending Gaussian elimination to bigger matrices.

1.2 Course Overview

The central focus is computing solutions of linear systems and eigenvalue problems for dense and large sparse matrices.



Figure 1.1: Sparsity pattern of a tridiagonal matrix.

1.2.1 The five main chapters

Chapter	Subject	Central ideas
1. Introduction	Applications and motivation	Discretized differential equations, least squares, sparse matrices, PageRank
2. Orthogonalization and QR Factorization	Constructing orthonormal bases	Gram–Schmidt, Householder transformations, Givens rotations
3. Dense Algebraic Eigenvalue Problems	Computing eigenvalues of dense matrices	Schur decomposition, eigenvalue conditioning, vector and subspace iterations, QR iteration
4. Large Sparse Linear Systems	Solving very large $Ax = b$ problems	Jacobi, Gauss–Seidel, projection methods, Krylov subspaces, CG, GMRES, preconditioning
5. Large Sparse Eigenvalue Problems	Finding selected eigenvalues of huge matrices	Rayleigh–Ritz, Lanczos, Arnoldi, implicitly restarted Arnoldi, interior eigenvalues

1.2.2 Relationship Between Chapters

orthogonalization and projection are the unifying ideas behind much of the course.

Orthogonalization provides suitable bases for subspaces. Projection reduces large computational problems to smaller ones.

A Krylov subspace has the form:

$$\mathcal{K}_m(A, v) = \text{span}\{v, Av, A^2v, \dots, A^{m-1}v\}.$$

This same construction appears in both large sparse linear-system solvers an dlarge sparse eigensovles. The same Arnoldi or Lanczos machinery can support both kinds of computation.

1.3 Linear Algebra and Numerics Prerequisites and Review

1.3.1 Essential linear algebra topics

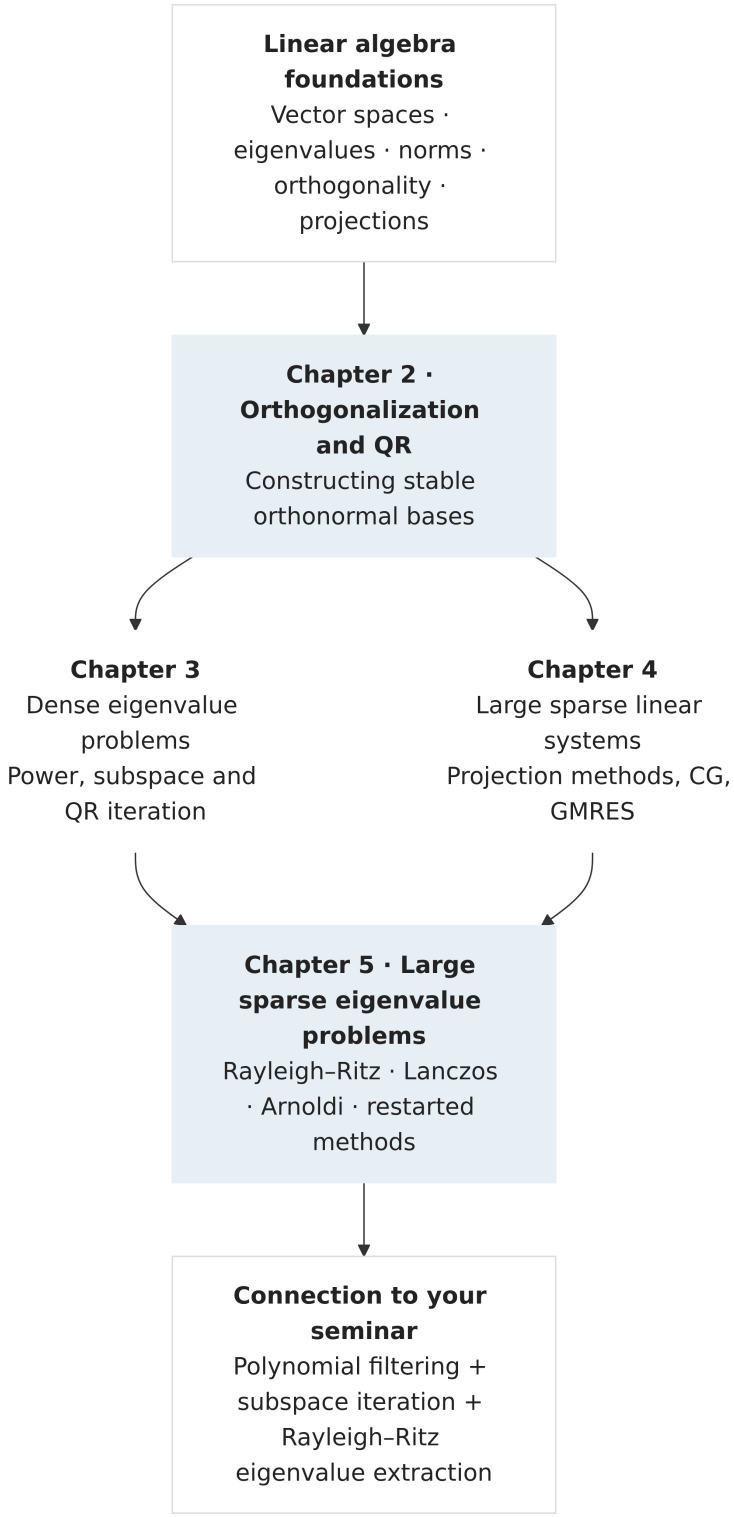


Figure 1.2: Conceptual relationships between the main chapters.

Topic	What you should be able to do	Why it matters for NLA
Vector spaces, bases and dimension	Work with span, linear independence, rank, kernel, image and basis changes	Almost every iterative algorithm constructs or manipulates subspaces
Inner products and orthogonality	Calculate inner products, orthogonal complements, orthonormal bases	Gram–Schmidt, QR, Arnoldi, Lanczos
Orthogonal projections	Understand projections onto subspaces and their matrix representation	Least squares, Galerkin and Rayleigh–Ritz methods
Eigenvalues and eigenvectors	Compute simple eigenpairs, understand diagonalization and eigenspaces	Chapters 3 and 5, and your seminar
Symmetric and Hermitian matrices	Understand the spectral theorem and orthogonal diagonalization	CG, Lanczos, eigenvalue conditioning
Positive-definite matrices	Recognize SPD matrices and understand quadratic forms	Cholesky, CG, energy minimization
Similarity transformations	Interpret $V^{-1}AV$ as a basis change	Schur decomposition, QR eigenvalue iteration
Matrix norms and spectral radius	Understand operator norms, eigenvalue magnitudes and basic inequalities	Error estimates, convergence, conditioning

Especially three concepts deserve extra attention:

Orthogonality, projection and subspaces

Suppose $Q \in \mathbb{R}^{n\times m}$ has orthonormal columns:

$$Q^TQ = I_m.$$

Then the matrix

$$P = QQ^T$$

is the orthogonal projector onto the subspace spanned by the columns of Q.

Eigenvalues, diagonalization and the spectral theorem

For a general diagonalizable matrix,

$$A = V\Lambda V^{-1}.$$

For a real symmetric matrix, the much stronger result

$$A = Q\Lambda Q^T, \qquad Q^TQ = I.$$

Is the spectral theorem. It t ells us that a symmetric matrix has real eigenvalues and admits an orthonormal basis or eigenvectors.

Consider an inital vector

$$v = \sum_{i=1}^{n} \alpha_i q_i,$$

where q_i are the orthonormal eigenvectors of A. Then

$$A^k v = \sum_{i=1}^{n} \alpha_i \lambda_i^k q_i.$$

Applying the matrix repeatedly changes the relative importance of the eigenvector accordidng to the magnitued of their eigenvalues. This explains the fundamental mechanism of **power iteration** and, after generalization, subspace iteration and polynomial filtering.

Symmetric positive definiteness and quadratic forms

A real symmetric matrix A is positive definite when

$$x^TAx > 0 \qquad \text{for every } x \neq 0.$$

Equivalent characterizations include the positivity of all eigenvalues and the existence of a Cholesky factorization.

The relationship between SPD matrices and minimization problems is especially useful. For example, solving $Ax = b$ for an SPD matrix is equivalnet to minimizing

$$f(x) = \frac{1}{2}x^TAx - b^Tx.$$

This provides a natural connection between linear systems, quadratic optimization and the conjugate gradient method.

1.3.2 Prerequisites from Numerics 0

Subject	Bastian's `num0.pdf`	Scheichl–Zech's `Num0_notes_ss25.pdf`	Priority
Floating-point arithmetic and rounding errors	§§2–3	§§2.3–2.4	High
Conditioning, matrix norms and stability	§§3, 5–6	§§2.1–2.2, 3.1–3.3	Very high
Direct methods: Gaussian elimination, LU, Cholesky	§§7–10	§§3.4–3.7	Medium–high
Least squares, QR and Householder	§§11–12	§§3.8–3.9	Very high
Basic iterative linear solvers	§§13–14	§6.3	Very high
Polynomial approximation, especially Chebyshev	§§17–19, 23 for broader background	§§4.9–4.10	High for seminar

Conditioning versus stability

Conditioning is a property of a mathematical problem. Stability is a property of an algorithm used to solve that problem.

For instance, suppose $Ax = b$ and we perturb the right-hand side by Δb. The solution changes according to

$$\Delta x = A^{-1}\Delta b$$

The condition number

$$\kappa(A) = \|A\|\|A^{-1}\|$$

measures the potential amplification of relative errors. But even if a problem is well-conditioned, a poor algorithm can introduce unnecessarily large computational errors.

It is alos essential to distinguish the residual

$$r = b - A\tilde{x}$$

from the actual solution error

$$e = x - \tilde{x}.$$

They are related by

$$Ae = r, \qquad e = A^{-1}r.$$

Thus a small residual does not automatically guarantee a small forward error when the matrix is ill-conditioned.

These ideas reappear throghout NLA, including in the conditioning of eigenvalues and eigenvectors.

Matrix factorizations and orthogonal transformations

Distinction between QR and LU:

- LU decomposes a matrix into triangular factors and supports the direct solution of linear systems.
- QR constructs orthonormal bases and triangular factors, which are especially useful least squares and eigenvalue algorithms.
- Cholesky exploits symmetric positive definiteness.

For NLA **QR is more urgent than detailed LU pivoting analysis**, because Ch 2 of NLA directly develops Gram-Schmidt, Householder and Givens transformations.

Basic iterative methods and convergence

Consider an iteration

$$x^{(k+1)} = Bx^{(k)} + c.$$

If x^* is its fixed point, then the error satisfies

$$e^{(k+1)} = Be^{(k)}.$$

Consequently,

$$e^{(k)} = B^k e^{(0)}.$$

The fundamental convergence criterion is

$$\boxed{\rho(B) < 1}$$

where $\rho(B)$ is the spectral radius.

This connects eigenvalues directly to the convergence of numerical algorithms. Jacobi and Gauss-Seidel should be revised, but with a focus on understanding how their iteration matrices arise and how their convergence can be analyzed.

Chevyshev Polynomials

The NLA script has Appendix B.2: Chebyshev polynomials, beginning on printed page 153.

It introduces the recurrence

$$T_0(x) = 1, \qquad T_1(x) = x,$$

$$T_{k+1}(x) = 2xT_k(x) - T_{k-1}(x),$$

and properties of Chebyshev polynomials on and outside $[-1, 1]$

1.3.3 The connection to the Chebyshev-filtered subspace iteration seminar

Ordinary subspace iteration

Suppose A is symmetric and we wish to compute several of its dominant eigenpairs.

We begin with a matrix Q_0 whose columns are orthonormal and repeatedly compute

$$Y_{k+1} = AQ_k,$$

followed by orthonormalization,

$$Q_{k+1} = \text{orth}(Y_{k+1}).$$

The successive subspaces approach an invariant subspace corresponding to dominant eigenvalues, subject to appropriate initial-subspace and spectral-gap assumptions.

Polynomial-filtered iteration

Instead of multiplying by A, we apply a polynomial in A:

$$Y_{k+1} = p_m(A)Q_k.$$

Why does this help?

Because for an eigenvector q_i,

$$Aq_i = \lambda_i q_i \quad \Longrightarrow \quad p_m(A)q_i = p_m(\lambda_i)q_i.$$

We can therefore choose a polynomial that makes the desired eigenvalue components comparatively large and suppresses the unwanted components.

A Chebyshev polynomial, after suitable shifting and scaling, is an effective choice because of its controlled magnitude on a specified interval and rapid growth outside it.

The important point is that we can apply such a polynomial using a recurrence involving matrix–vector products, without constructing the generally dense matrix $p_m(A)$.

What is needed to understand the complete algorithm

Component	Mathematical prerequisite	NLA connection
Spectral transformation $p(A)$	Diagonalization, matrix polynomials	Appendix A.1.2, §3.3
Desired invariant subspace	Subspaces, eigenspaces, spectral gaps	§§3.2–3.3
Chebyshev filter	Polynomial recurrence and approximation	Appendix B.2
Orthonormalization	Inner products, Gram–Schmidt, QR	Chapter 2
Subspace update	Orthogonal subspace iteration	§3.3.2
Approximate eigenpairs	Rayleigh quotient, orthogonal projection	§5.1
Competing eigensolvers	Krylov spaces, Lanczos, Arnoldi	Chapters 4–5

The script even develops polynomial filtering in §3.3. There is an example using

$$A = \mathrm{diag}(1, 2, 3)$$

and the polynomial

$$p(t) = (t-1)(t-3).$$

Here the unwanted eigenvalues 1 and 3 are mapped to zero, while the component associated with eigenvalue 2 is retained.

Chapter 5 then combines projection, Rayleigh–Ritz approximation and Krylov-based eigenvalue methods. This is a natural starting point for comparing your seminar algorithm with Lanczos and Arnoldi methods.

One subtle but important distinction is that the seminar's polynomial-filtered subspace method need not itself be a Krylov-subspace algorithm in the same sense as Arnoldi. Both make essential use of matrix polynomials and subspaces, but organize their iterations differently.

1.4 Revision Strategy

1.4.1 Phase 1 - linear algebra foundation

Vector spaces and bases, innder products, projections, eigenvalues, diagonalization, and SPD matrices.

Useful readiness test:

> can you explain, and then prove why QQ^T is an orthogonal projector when Q has orthonormal columns

1.4.2 Phase 2 - numerical-analysis foundation

Conditioning, residuals and error estimates, floating-point arithmetic, QR factorization and convergence of iterative methods.

readiness test:

1. why does conditioning matter?
2. why can a small residual be misleading?
3. why does the spectral radius control stationary iterations?

1.4.3 Phase 3 - Eigenvalues

Power method, Rayleigh quitient, subspace iteration and polynonial filtering. Chebyshev polynomials and Rayleigh-Ritz.

1.4.4 Targeted Revision Plan

11 focused sessions.

Week 1 - Rebuild the linear algebra foundations

Goal: Recover fluency with vectors, subspaces, projections, and orthogonalization so that Chapter 2 becomes manageable.

Session	Material to revise	Mastery check
1. Vector spaces and linear maps · 60 min	Span, linear independence, bases, dimension, rank, kernel, image, change of basis	Given a matrix, determine bases for its kernel and image; explain rank–nullity
2. Inner products and orthogonality · 75 min	Inner products, orthogonal complements, orthonormal bases, Gram–Schmidt	Construct an orthonormal basis from linearly independent vectors
3. Orthogonal projections · 75 min	Projection onto a line/subspace, $P = QQ^T$, least-squares geometry	Derive the projection matrix and prove $P^2 = P$
4. QR factorization · 90 min	Gram–Schmidt, Householder reflections, QR, least-squares solution	Perform QR on a small matrix and explain why orthogonality is useful numerically

Reading sources: NLA Appendix A.1–A.3, NLA Chapter 2, Bastian's §5 and §§11–12, and Scheichl–Zech's §§3.8–3.9.

Week 2 - Recover spectral theory and numerical analysis

Goal: Understand how matrix structure, conditioning and numerical algorithms interact.

Session	Material to revise	Mastery check
5. Eigenvalues and the spectral theorem · 90 min	Eigenvectors, eigenspaces, diagonalization, symmetric/Hermitian matrices, SPD	Diagonalize a symmetric matrix and use the decomposition to compute $A^k v$
6. Matrix norms and conditioning · 75 min	Induced norms, spectral radius, condition numbers, residual versus error, stability	Derive an error estimate using $e = A^{-1}r$ and explain why small residuals can mislead

Session	Material to revise	Mastery check
7. Direct numerical methods · 60 min	Gaussian elimination, LU, pivoting, triangular systems, Cholesky	Solve a system via LU and explain when Cholesky applies
8. Basic iterative methods · 75 min	Fixed-point iteration, Jacobi, Gauss–Seidel, iteration matrices, convergence	Derive $e^{(k+1)} = Be^{(k)}$ and explain the role of $\rho(B)$

Reading sources: NLA §§3.1–3.2, Appendix B.1 and §4.2; Bastian's §§3, 5–10 and 13–14; Scheichl–Zech's §§2–3 and 6.3.

Week 3 - Connect the foundations to eigensolvers and your seminar

Goal: Build a conceptual understanding of polynomial-filtered subspace iteration.

Session	Material to revise	Mastery check
9. Power and subspace iteration · 90 min	Power method, spectral gaps, invariant subspaces, orthogonal subspace iteration	Derive why the power method favors a dominant eigenvector and explain why subspace iteration requires orthonormalization
10. Matrix polynomials and Chebyshev filtering · 90 min	$p(A)$, spectral mapping for polynomials, Chebyshev recurrence, interval scaling	Explain why applying a polynomial to A can suppress unwanted eigenvector components
11. Rayleigh–Ritz and integration · 75 min	Q^TAQ, Ritz values/vectors, residual orthogonality, connection to filtered subspace iteration	Derive the Rayleigh–Ritz projection and describe a complete filtered eigensolver at a high level

Reading sources: NLA §§3.3, 5.1 and Appendix B.2; Scheichl–Zech's §4.10 for deeper Chebyshev approximation background.

The NLA script explicitly introduces polynomial filtering within vector iteration and later Rayleigh–Ritz approximation for large sparse eigenvalue problems.

Specific exercises to verify your pogress

Rather than completing a large number of routine exercises, a small nunmber that requires to combing computation with explanation

For examle five representative checkpoint problems

- **Checkpoint A - Linear algebra**

Given

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix},$$

determine $\ker(A)$, $\mathrm{im}(A)$, rank and nullity. Relate your results to invertibility.

- **Checkpoint B - Orthogonal projections**

Let

$$q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Compute $P = qq^T$, prove that P is an orthogonal projector, and use it to project $(1, 0)^T$.

- **Checkpoint C - Spectral theorem**

Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Determine its eigenvalues and an orthonormal eigenbasis. Express A^k in terms of its eigendecomposition.

- **Checkpoint D - Numerical Conditioning**

Consider

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 10^{-4} \end{pmatrix}.$$

Calculate $\kappa_2(A)$. Explain how small perturbations in the second component of b can affect the solution of $Ax = b$.

- **Checkpoint D - Numerical Conditioning**

Consider

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 10^{-4} \end{pmatrix}.$$

Calculate $\kappa_2(A)$. Explain how small perturbations in the second component of b can affect the solution of $Ax = b$.

- **Checkpoint E - Polynomial Filtering**

Let

$$A = \operatorname{diag}(1, 2, 3), \qquad p(t) = (t-1)(t-3).$$

Determine $p(A)$. Explain how repeated application of $p(A)$ affects each eigenspace.

The first four checkpoints test prerequisites. The last one connects directly to the polynomial-filtering example in NLA §3.3.

References

Part I

Review

2 Session 1 — Vector Spaces and Linear Maps

The central idea of this session is that **a linear map acts on vectors, while a matrix describes that action in chosen coordinates**. Bases connect these two viewpoints. Kernel, image, rank, and nullity describe what the map loses and what it can produce.

Allow approximately **35–40 minutes for the notes and worked examples**, followed by **20–25 minutes for the exercises**. The aim is to reconstruct the reasoning, not simply read the formulas.

Throughout, vector spaces are finite-dimensional over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, and coordinate vectors are columns.

2.1 Subspaces and span

A **vector space** is a set whose elements can be added and multiplied by scalars, obeying the usual distributive and associative laws, with a zero vector and additive inverses. Examples include $\mathbb{R}^n$, $\mathbb{C}^n$, and the space

$$\mathcal{P}_2 = \{a + bt + ct^2 : a, b, c \in \mathbb{F}\}$$

of polynomials of degree at most two.

The important point is that a "vector" need not be an arrow: it can be a polynomial, a function, or a matrix.

2.1.1 Subspaces

A subset $U \subseteq V$ is a **subspace** if it is itself a vector space under the operations inherited from V.

To check this, it suffices to verify

$$0 \in U, \qquad \alpha u + \beta v \in U \quad \text{for all } u, v \in U,\ \alpha, \beta \in \mathbb{F}.$$

Thus a subspace is closed under linear combinations. Over $\mathbb{R}$, lines and planes through the origin are familiar examples.

2.1.2 Span

The **span** of vectors $v_1, \dots, v_k$ is

$$\operatorname{span}\{v_1, \dots, v_k\} = \left\{ \sum_{j=1}^{k} c_j v_j : c_j \in \mathbb{F} \right\}.$$

It consists of all vectors obtainable by combining the given vectors. It is the smallest subspace containing them: any subspace containing the generators must also contain all their linear combinations.

2.1.3 Worked example: generators and constraints

Let

$$u = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \qquad v = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \qquad w = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}.$$

Because $w = u + v$,

$$\operatorname{span}\{u, v, w\} = \operatorname{span}\{u, v\}.$$

Moreover,

$$\alpha u + \beta v = \begin{pmatrix} \alpha \\ \beta \\ \alpha + \beta \end{pmatrix},$$

so this span is exactly

$$U = \left\{ x \in \mathbb{R}^3 : x_3 = x_1 + x_2 \right\}.$$

We have described the same plane in two ways:

- **By generators:** $U = \operatorname{span}\{u, v\}$.
- **By a homogeneous constraint:** $x_3 - x_1 - x_2 = 0$.

Replacing the constraint by $x_3 - x_1 - x_2 = 1$ gives a translated plane. It is not a subspace because it does not contain zero.

2.2 Independence, bases, and dimension

Span answers: **Which vectors can we represent?**

Linear independence answers: **Is the representation unique?**

2.2.1 Linear independence

A family $v_1, \dots, v_k$ is **linearly independent** if

$$\sum_{j=1}^{k} c_j v_j = 0 \quad \Longrightarrow \quad c_1 = \cdots = c_k = 0.$$

Otherwise it is **linearly dependent**.

A nontrivial dependence relation allows at least one vector to be expressed using the others. Indeed, if $c_\ell \neq 0$, then

$$v_\ell = -\sum_{j \neq \ell} \frac{c_j}{c_\ell} v_j.$$

Thus dependence means redundancy. It does not require two of the vectors to be parallel.

2.2.2 Bases and coordinates

An ordered family

$$\mathcal{B} = (b_1, \dots, b_n)$$

is a **basis** of V if it is independent and spans V.

Every $x \in V$ then has a unique representation

$$x = \xi_1 b_1 + \cdots + \xi_n b_n.$$

Its coordinate vector relative to $\mathcal{B}$ is

$$[x]_{\mathcal{B}} = \begin{pmatrix} \xi_1 \\ \vdots \\ \xi_n \end{pmatrix}.$$

Why unique? If two coefficient lists represent x, subtracting the representations gives a zero linear combination of the basis vectors. Independence forces the corresponding coefficients to agree.

Consequently:

> **Spanning guarantees existence of coordinates; independence guarantees uniqueness.**

In the preceding example, u, v are independent because

$$\alpha u + \beta v = 0$$

forces $\alpha = \beta = 0$ from the first two components. Hence (u, v) is a basis of U, and

$$\left[\begin{pmatrix} a \\ b \\ a+b \end{pmatrix} \right]_{(u,v)} = \begin{pmatrix} a \\ b \end{pmatrix}.$$

The vector has three ambient components but only two coordinates within the plane.

2.2.3 Dimension

Theorem. Every basis of a finite-dimensional vector space has the same number of elements. This number is its **dimension**.

Two related facts are particularly useful:

1. Every independent family can be extended to a basis.
2. Every finite spanning family contains a basis.

For the first, repeatedly append a vector outside the current span. For the second, repeatedly remove a redundant vector. The underlying counting principle is that an independent family cannot contain more vectors than a spanning family; applying this to two bases in both directions proves that their lengths agree.

Therefore, in a space of dimension n:

- Any n independent vectors form a basis.
- Any n spanning vectors form a basis.
- More than n vectors are dependent.
- Fewer than n vectors cannot span.

The zero space $\{0\}$ has dimension zero and the empty basis. The family consisting of the zero vector is not independent.

2.2.4 A basis stored as a matrix

For vectors $b_j \in \mathbb{F}^m$, write

$$B = \begin{pmatrix} b_1 & \cdots & b_k \end{pmatrix} \in \mathbb{F}^{m \times k}.$$

Then

$$Bc = \sum_{j=1}^{k} c_j b_j.$$

Thus matrix–vector multiplication constructs a vector from its coefficients.

If the columns are independent, they form a basis of their column span, and

$$x = B[x]_{\mathcal{B}} \qquad (x \in \operatorname{span}\{b_1, \dots, b_k\}).$$

A basis matrix for a subspace can be rectangular. It need not possess a two-sided inverse.

2.3 Linear maps and their matrix representations

A map $T : V \to W$ is **linear** if

$$T(\alpha x + \beta y) = \alpha T(x) + \beta T(y).$$

In particular, $T(0) = 0$.

Linearity implies that a map is completely determined by its values on a basis:

$$T\left(\sum_j \xi_j b_j\right) = \sum_j \xi_j T(b_j).$$

2.3.1 Constructing the representing matrix

Choose bases

$$\mathcal{B} = (b_1, \dots, b_n) \quad \text{of } V, \qquad \mathcal{C} = (c_1, \dots, c_m) \quad \text{of } W.$$

The matrix

$$A = [T]_{\mathcal{C} \leftarrow \mathcal{B}}$$

is defined by taking the coordinates of $T(b_j)$ as its j-th column:

$$A = \begin{pmatrix} [T(b_1)]_{\mathcal{C}} & \cdots & [T(b_n)]_{\mathcal{C}} \end{pmatrix}.$$

Then

$$\boxed{[T(x)]_{\mathcal{C}} = A[x]_{\mathcal{B}}.}$$

This is the fundamental relationship:

The map takes vectors to vectors; its matrix takes input coordinates to output coordinates.

In standard bases, Ae_j is simply the j-th column of A.

2.3.2 Worked example: differentiation

Consider

$$D : \mathcal{P}_2 \longrightarrow \mathcal{P}_1, \qquad D(p) = p'.$$

Choose

$$\mathcal{B} = (1, t, t^2), \qquad \mathcal{C} = (1, t).$$

Since

$$D(1) = 0, \qquad D(t) = 1, \qquad D(t^2) = 2t,$$

we obtain

$$[D]_{\mathcal{C} \leftarrow \mathcal{B}} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Indeed,

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} b \\ 2c \end{pmatrix},$$

which represents the derivative of $a + bt + ct^2$.

The matrix has three columns because the domain has dimension three, and two rows because the codomain has dimension two.

2.4 Kernel, image, and solvability

For a linear map $T : V \to W$, define

$$\ker T = \{x \in V : T(x) = 0\},$$

and

$$\operatorname{im} T = \operatorname{ran} T = \{T(x) : x \in V\}.$$

The **kernel** consists of input directions erased by the map. The **image**, also called the **range**, consists of attainable outputs.

Both are subspaces. For example, if $u, v \in \ker T$, then

$$T(\alpha u + \beta v) = \alpha T(u) + \beta T(v) = 0.$$

For the image,

$$\alpha T(u) + \beta T(v) = T(\alpha u + \beta v)$$

is again an attainable output.

Define

$$\operatorname{nullity} T = \dim \ker T, \qquad \operatorname{rank} T = \dim \operatorname{im} T.$$

For $A \in \mathbb{F}^{m\times n}$, regarded as the map $x \mapsto Ax$,

$$\ker A \subseteq \mathbb{F}^n, \qquad \operatorname{im} A \subseteq \mathbb{F}^m.$$

Since $Ax = \sum_j x_j a_j$,

$$\operatorname{im} A = \operatorname{span}\{a_1, \dots, a_n\}.$$

2.4.1 Solvability and all solutions

The equation $T(x) = b$ is solvable exactly when

$$b \in \operatorname{im} T.$$

If x_p is one solution, then

$$\boxed{\{x : T(x) = b\} = x_p + \ker T.}$$

Proof.

$$T(x) = b = T(x_p) \iff T(x - x_p) = 0 \iff x - x_p \in \ker T.$$

The kernel therefore describes the ambiguity in reconstructing an input from an output. A nonempty solution set is a translate of the kernel; for $b \neq 0$, it is not a subspace.

2.4.2 Injectivity and surjectivity

A map is **injective** if equal outputs imply equal inputs. Because

$$T(x) = T(y) \iff x - y \in \ker T,$$

we have

$$\boxed{T \text{ injective} \iff \ker T = \{0\}.}$$

A map is **surjective** if every vector in its codomain is attained:

$$\boxed{T \text{ surjective} \iff \operatorname{im} T = W.}$$

For differentiation,

$$\ker D = \operatorname{span}\{1\}, \qquad \operatorname{im} D = \mathcal{P}_1.$$

Thus $D : \mathcal{P}_2 \to \mathcal{P}_1$ is surjective but not injective: differentiation loses the constant term.

For instance,

$$p' = 2 + 6t \quad \Longleftrightarrow \quad p(t) = 2t + 3t^2 + c.$$

2.4.3 Worked example: a complete rectangular system

Consider

$$A = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 2 & 1 & 2 \end{pmatrix}, \qquad A : \mathbb{R}^4 \to \mathbb{R}^3.$$

Subtracting rows 1 and 2 from row 3 gives

$$R = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The pivots lie in columns 1 and 3. Variables x_2, x_4 are free.

Kernel. Set $x_2 = s$, $x_4 = t$. The homogeneous equations give

$$x_1 = -2s - t, \qquad x_3 = -t.$$

Hence

$$x = s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ -1 \\ 1 \end{pmatrix}.$$

These vectors span the kernel and are independent: their second and fourth components recover s, t. Therefore

$$\ker A = \operatorname{span} \left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ -1 \\ 1 \end{pmatrix} \right\}, \qquad \text{nullity } A = 2.$$

Image. Take the pivot columns of the **original matrix**:

$$a_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \qquad a_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

They are independent, and

$$a_2 = 2a_1, \qquad a_4 = a_1 + a_3.$$

Thus

$$\operatorname{im} A = \operatorname{span}\{a_1, a_3\} = \{b \in \mathbb{R}^3 : b_3 = b_1 + b_2\}, \qquad \operatorname{rank} A = 2.$$

Solvability. Applying the row operation to the augmented matrix gives

$$\left(\begin{array}{cccc|c} 1 & 2 & 0 & 1 & b_1 \\ 0 & 0 & 1 & 1 & b_2 \\ 0 & 0 & 0 & 0 & b_3 - b_1 - b_2 \end{array}\right).$$

Therefore,

$$Ax = b \text{ is solvable } \iff b_3 = b_1 + b_2.$$

For $b = (3, 2, 5)^T$, one solution is $x_p = (3, 0, 2, 0)^T$, and every solution is

$$x = \begin{pmatrix} 3 \\ 0 \\ 2 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ -1 \\ 1 \end{pmatrix}.$$

For $b = (3, 2, 6)^T$, no solution exists.

2.4.4 Why use the original pivot columns?

Row operations produce $R = EA$, where E is invertible. Consequently,

$$Rx = 0 \iff Ax = 0,$$

so the kernel is unchanged.

They also preserve relations among columns:

$$\sum_j c_j E a_j = 0 \iff \sum_j c_j a_j = 0.$$

Thus pivot-column indices identify independent spanning columns in the original matrix.

However,

$$\operatorname{im} R = E(\operatorname{im} A),$$

which is generally a different subspace. Here $\operatorname{im} R$ is the plane $b_3 = 0$, while $\operatorname{im} A$ is the plane $b_3 = b_1 + b_2$.

2.5 The rank–nullity theorem

Theorem. For a linear map $T : V \to W$,

$$\boxed{\dim \ker T + \dim \operatorname{im} T = \dim V.}$$

2.5.1 Proof by extending a kernel basis

Let $k_1, \dots, k_q$ be a basis of $\ker T$. Extend it to a basis of V:

$$(k_1, \dots, k_q, v_1, \dots, v_{n-q}), \qquad n = \dim V.$$

We show that

$$T(v_1), \dots, T(v_{n-q})$$

form a basis of the image.

They span the image. Every input can be expanded in the extended basis. Applying T annihilates all kernel terms, leaving a combination of the $T(v_j)$.

They are independent. Suppose

$$\sum_j \alpha_j T(v_j) = 0.$$

Then

$$\sum_j \alpha_j v_j \in \ker T,$$

so it equals some $\sum_i \beta_i k_i$. Therefore

$$\sum_j \alpha_j v_j - \sum_i \beta_i k_i = 0.$$

Independence of the extended basis forces every coefficient to vanish.

Hence

$$\dim \operatorname{im} T = n - q,$$

which proves the theorem.

2.5.2 Interpretation and consequences

There are q independent input directions that the map erases. The remaining $n - q$ basis directions produce independent outputs.

This is a dimension statement. It does not say that kernel and image belong to the same ambient space or are orthogonal.

For the preceding 3×4 example,

$$\operatorname{rank} A + \operatorname{nullity} A = 2 + 2 = 4,$$

the number of **columns**, which is the domain dimension.

For $A \in \mathbb{F}^{m \times n}$, with rank r:

Property	Equivalent condition	Meaning for $Ax = b$
Injective	$r = n$, full column rank	At most one solution for each b
Surjective onto $\mathbb{F}^m$	$r = m$, full row rank	At least one solution for every b
Bijective	$m = n = r$	Exactly one solution for every b

Since $r \leq \min(m, n)$, a wide matrix cannot be injective, and a tall matrix cannot be surjective onto its full codomain.

For a square matrix,

$$\text{injective} \iff \text{surjective} \iff \text{invertible}.$$

A consistent system has $n - r$ free parameters.

2.6 Change of basis

The vector stays the same when we change its basis; its coordinate vector changes.

Let $\mathcal{B}$ and $\widetilde{\mathcal{B}}$ be bases of the same space. Define

$$S = \begin{pmatrix} [\tilde{b}_1]_{\mathcal{B}} & \cdots & [\tilde{b}_n]_{\mathcal{B}} \end{pmatrix}.$$

Expanding a vector in the new basis and then expressing those basis vectors in the old basis gives

$$\boxed{[x]_{\mathcal{B}} = S[x]_{\widetilde{\mathcal{B}}}.}$$

Thus:

- S converts **new coordinates to old coordinates**.
- S^{-1} converts **old coordinates to new coordinates**.

Its columns are new basis vectors expressed in old coordinates.

2.6.1 Changing a map's matrix

Suppose

$$A = [T]_{\mathcal{C} \leftarrow \mathcal{B}},$$

and the new input and output bases satisfy

$$[x]_{\mathcal{B}} = S[x]_{\widetilde{\mathcal{B}}}, \qquad [y]_{\mathcal{C}} = P[y]_{\widetilde{\mathcal{C}}}.$$

Then

$$[T(x)]_{\widetilde{\mathcal{C}}} = P^{-1}[T(x)]_{\mathcal{C}} = P^{-1}AS[x]_{\widetilde{\mathcal{B}}}.$$

Therefore,

$$\boxed{\widetilde{A} = P^{-1}AS.}$$

Read the product from right to left:

1. Convert input coordinates to the old basis.
2. Apply the old representing matrix.
3. Convert output coordinates to the new basis.

For an endomorphism $T : V \to V$, if we use the same basis change for input and output, $P = S$, giving

$$\boxed{\widetilde{A} = S^{-1}AS.}$$

This is a **similarity transformation**.

2.6.2 Worked example: coordinates reveal a simpler action

In standard coordinates, let

$$T(x) = Ax, \qquad A = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}.$$

Choose

$$b_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad b_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

The basis matrix and its inverse are

$$S = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \qquad S^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Hence

$$S^{-1}AS = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}.$$

We can see this directly:

$$T(b_1) = 2b_1, \qquad T(b_2) = b_2.$$

These images have new coordinates $(2,0)^T$ and $(0,1)^T$, respectively.

For $x = (3,1)^T$,

$$[x]_{\mathcal{B}} = S^{-1}x = \begin{pmatrix} 4 \\ 1 \end{pmatrix}.$$

Applying the new matrix gives

$$[T(x)]_{\mathcal{B}} = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 8 \\ 1 \end{pmatrix}.$$

Converting back,

$$T(x) = S \begin{pmatrix} 8 \\ 1 \end{pmatrix} = \begin{pmatrix} 7 \\ 1 \end{pmatrix} = Ax.$$

The map has not changed; its action is simpler to describe in the new basis.

Invertible coordinate changes preserve rank and nullity. The coordinate descriptions of kernel and image transform according to

$$\ker(P^{-1}AS) = S^{-1}(\ker A), \qquad \operatorname{im}(P^{-1}AS) = P^{-1}(\operatorname{im} A).$$

2.7 Connections to numerical linear algebra

Let $V \in \mathbb{F}^{n\times k}$ have independent columns spanning a subspace U. If $M \in \mathbb{F}^{k\times k}$ is invertible, then

$$\operatorname{im}(VM) = \operatorname{im} V,$$

because Mc ranges over all of $\mathbb{F}^k$.

This distinction between a **subspace** and a **basis representing it** is fundamental.

Orthogonalization and QR. A thin QR factorization

$$V = QR$$

replaces independent columns of V by orthonormal columns of Q. Since R is invertible, the span is preserved. Moreover,

$$x = Vc = Q(Rc),$$

so R converts V-basis coordinates into Q-basis coordinates. Sessions 2 and 4 develop the geometry and algorithms.

Krylov subspaces.

$$\mathcal{K}_k(A, v) = \text{span}\{v, Av, \dots, A^{k-1}v\}.$$

There are k generators, but their span may have smaller dimension if they are dependent. Counting generated vectors is not the same as counting independent directions.

Subspace iteration and filtering. Applying A, or a polynomial

$$p(A) = \sum_{j=0}^{d} \alpha_j A^j,$$

to a basis matrix produces the image of its subspace under that map. This may change the subspace and lower its dimension. Subsequent orthogonalization preserves the resulting span; it cannot recover directions erased by the map.

Invariant subspaces and projection methods. A subspace U is invariant under A if

$$AU \subseteq U.$$

If V is a basis matrix of U, this is equivalent to

$$AV = VH$$

for a unique $k \times k$ matrix H. Its columns contain the coordinates of the vectors Av_j within U. Thus H represents the restriction of A to U. If U is not invariant, projection methods instead represent a projected action, studied later.

Finally, our calculations used **exact rank**. In floating-point arithmetic, nearly dependent columns can be difficult to distinguish from dependent ones. For example,

$$\begin{pmatrix} 1 & 0 \\ 0 & \varepsilon \end{pmatrix}$$

has exact rank two whenever $\varepsilon \neq 0$, however small. Numerical rank requires a tolerance and attention to scale.

2.8 Independent exercises

Attempt these before opening the solutions below.

2.8.1 Exercise 1 — Subspaces and coordinates

Let

$$U = \{(a, b, a+b)^T : a, b \in \mathbb{R}\}, \qquad F = \{(a, b, a+b+1)^T : a, b \in \mathbb{R}\}.$$

Justify that U is a subspace and F is not. Show that

$$\mathcal{B} = ((1, 1, 2)^T, (1, -1, 0)^T)$$

is a basis of U, and find $[(3, 1, 4)^T]_{\mathcal{B}}$.

2.8.2 Exercise 2 — A rectangular system

For

$$B = \begin{pmatrix} 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 1 \\ 1 & 2 & 1 & 3 \end{pmatrix},$$

find bases of its kernel and image, compute rank and nullity, and verify rank–nullity.

Characterize all $b \in \mathbb{R}^3$ for which $Bx = b$ is solvable, and find every solution for $b = (2, 1, 3)^T$.

Is this map injective or surjective? Explain why columns of its row-reduced form cannot automatically serve as an image basis of B.

2.8.3 Exercise 3 — Changing coordinates

Let

$$T(x_1, x_2)^T = (x_1 + 2x_2, 3x_2)^T.$$

Use the basis

$$\mathcal{B} = ((1, 0)^T, (1, 1)^T)$$

for both domain and codomain.

Compute the representing matrix both directly from basis images and by similarity. Verify your answer on $x = (0, 1)^T$.

2.8.4 Exercise 4 — Losing information in two stages

Let $T : V \to W$ and $L : W \to Z$ be linear. Prove

$$\ker T \subseteq \ker(LT), \qquad \operatorname{rank}(LT) \leq \min(\operatorname{rank} T, \operatorname{rank} L).$$

Explain how composition can lose directions, and give an example where $T, L \neq 0$ but $LT = 0$.

2.8.5 Exercise 5 — A map on polynomials

Define

$$E : \mathcal{P}_2 \to \mathbb{R}^2, \qquad E(p) = (p(0), p(1))^T.$$

Find its matrix in the input basis $(1, t, t^2)$ and the standard output basis. Determine its kernel, image, rank, and nullity, and describe every polynomial satisfying $p(0) = a$, $p(1) = b$.

Finally, use the input basis

$$(1 - t,\ t,\ t(t-1)).$$

Find the new matrix directly and explain why this basis is adapted to the interpolation problem.

2.9 Solutions — consult after attempting the exercises

2.9.1 Solution 1

The zero vector belongs to U, and linear combinations preserve the condition $x_3 = x_1 + x_2$. Thus U is a subspace. The zero vector does not belong to F, so F is not.

For $b_1 = (1, 1, 2)^T$, $b_2 = (1, -1, 0)^T$,

$$c_1 b_1 + c_2 b_2 = (c_1 + c_2, c_1 - c_2, 2c_1)^T.$$

Every $(a, b, a + b)^T \in U$ has the unique coefficients

$$c_1 = \frac{a+b}{2}, \qquad c_2 = \frac{a-b}{2}.$$

Therefore (b_1, b_2) is a basis, and

$$[(3, 1, 4)^T]_{\mathcal{B}} = (2, 1)^T.$$

2.9.2 Solution 2

Row reduction gives

$$B \longrightarrow R = \begin{pmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

The homogeneous equations are

$$x_1 = x_3 - x_4, \qquad x_2 = -x_3 - x_4.$$

Thus a kernel basis is

$$\left\{ \begin{pmatrix} 1 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

The third and fourth components establish independence.

The original pivot columns

$$c_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \qquad c_2 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$$

form an image basis: they are independent, and the other columns satisfy

$$c_3 = c_2 - c_1, \qquad c_4 = c_1 + c_2.$$

Hence rank and nullity are both two, and their sum is four.

The image is

$$\{b : b_3 = b_1 + b_2\}.$$

Indeed,

$$(b_1, b_2, b_1 + b_2)^T = (b_1 - b_2)c_1 + b_2 c_2.$$

For $b = (2, 1, 3)^T$, all solutions are

$$x = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} 1 \\ -1 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -1 \\ -1 \\ 0 \\ 1 \end{pmatrix}.$$

The map is neither injective nor surjective onto $\mathbb{R}^3$. Its kernel is nontrivial, and its image has dimension two.

The reduced matrix has image $\{y : y_3 = 0\}$, different from the original image. Row operations preserve column dependence relations, but generally move the column space.

2.9.3 Solution 3

Directly,

$$T(1,0)^T = (1,0)^T = b_1, \qquad T(1,1)^T = (3,3)^T = 3b_2.$$

Therefore

$$[T]_{\mathcal{B}\leftarrow\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}.$$

Alternatively,

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad S^{-1} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix},$$

and multiplication gives

$$S^{-1}AS = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}.$$

For $x = (0,1)^T$,

$$[x]_{\mathcal{B}} = (-1,1)^T, \qquad [T(x)]_{\mathcal{B}} = (-1,3)^T.$$

Converting back yields

$$S(-1,3)^T = (2,3)^T = Ax.$$

2.9.4 Solution 4

If $x \in \ker T$, then $L(Tx) = L(0) = 0$. Thus

$$\ker T \subseteq \ker(LT).$$

Also,

$$\operatorname{im}(LT) = L(\operatorname{im} T) \subseteq \operatorname{im} L,$$

so $\operatorname{rank}(LT) \leq \operatorname{rank} L$.

If $w_1, \ldots, w_r$ is an image basis of T, then

$$L(w_1), \ldots, L(w_r)$$

span the image of LT. Its dimension is therefore at most $r = \operatorname{rank} T$.

For example,

$$T = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad L = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

are both nonzero, but $LT = 0$. The first map's entire image lies in the second map's kernel.

2.9.5 Solution 5

The basis images are

$$E(1) = (1,1)^T, \qquad E(t) = (0,1)^T, \qquad E(t^2) = (0,1)^T,$$

so

$$[E] = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix}.$$

For $p = c_0 + c_1 t + c_2 t^2$, the kernel conditions give

$$c_0 = 0, \qquad c_1 + c_2 = 0.$$

Hence

$$\ker E = \operatorname{span}\{t^2 - t\}.$$

The first two matrix columns are independent, so

$$\operatorname{im} E = \mathbb{R}^2, \qquad \operatorname{rank} E = 2, \qquad \operatorname{nullity} E = 1.$$

One polynomial with the prescribed values is $a(1-t) + bt$. Every solution is therefore

$$\boxed{p(t) = a(1-t) + bt + c\,t(t-1), \qquad c \in \mathbb{R}.}$$

The new input basis has images

$$E(1-t) = (1,0)^T, \quad E(t) = (0,1)^T, \quad E(t(t-1)) = 0,$$

giving

$$[E]_{\text{standard} \leftarrow \widetilde{\mathcal{B}}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

To check that it is a basis, evaluate a vanishing linear combination at 0 and 1, forcing its first two coefficients to vanish; the third then vanishes because $t(t-1)$ is a nonzero polynomial.

This basis separates the two prescribed values from the one undetermined direction. Only the input basis changed, so this is not a similarity transformation.

2.10 End-of-session mastery checklist

Without consulting the notes, you should be able to:

- State the subspace test and distinguish a subspace from a translated solution set.
- Explain how spanning and independence provide existence and uniqueness of coordinates.
- Distinguish a vector, its coordinate vector, and a basis matrix.
- Construct a map's matrix from the images of basis vectors.
- Find a kernel basis using free variables and an image basis using original pivot columns.
- Characterize solvability by $b \in \operatorname{im} A$ and all solutions by $x_p + \ker A$.
- Prove rank–nullity by extending a kernel basis.
- Connect rank and nullity to injectivity, surjectivity, and invertibility.
- Derive $\widetilde{A} = P^{-1}AS$ and explain when it becomes similarity.
- Explain why changing a basis preserves a subspace, while applying a linear map may change its dimension.

Ready for Session 2: you can complete Exercises 2 and 3 independently, reconstruct the rank–nullity proof, and explain why a basis need not be orthogonal. Session 2 will add lengths, angles, and orthogonality to this algebraic foundation.